

Data and Artificial Intelligence

Cyber Shujaa Program

Week 4 Assignment Business Intelligence on Power BI (Hotel Management)

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1. Introduction

This assignment focused on developing hands-on experience in Business Intelligence (BI) using Microsoft Power BI. The exercise involved analyzing hotel management data to support business decision-making through data visualization and reporting.

The task provided an opportunity to apply BI concepts, including data loading, data transformation, data modeling, DAX calculations, and dashboard design. It also emphasized the importance of building analytical solutions that convert raw data into meaningful insights for business growth.

By the end of this assignment, I was able to understand how data from different sources (e.g., bookings, rooms, hotels, customers) can be integrated into a structured model and visualized to monitor performance indicators such as occupancy rates, revenue trends, and cancellation patterns.

2. Objectives

The main objectives of this assignment were:

- To gain hands-on experience in using Power BI for data analysis and visualization.
- To understand the structure of hotel business data and the client's analytical needs.
- To load and clean data from multiple sources using Power Query.
- To create a Star Schema data model linking fact and dimension tables.
- To develop DAX (Data Analysis Expressions) for key performance measures.
- To build an interactive and visually appealing dashboard for business insights.
- To publish the dashboard online and include it in a personal portfolio for professional presentation.

3. Tasks Completed

Step 1: Understand the Business and Client Needs

Before working with Power BI, I reviewed the hotel management dataset to understand what business questions needed to be answered. The client (hotel management) required insights into:

- Total revenue generated by room types and locations.
- Occupancy trends over time.
- Monthly booking patterns and seasonal demand.
- Cancellation rates and their impact on revenue.
- Customer and hotel performance metrics.

This initial understanding helped in designing a dashboard that supports decision-making in marketing, pricing, and operations management.

Step 2: Load and Transform Data

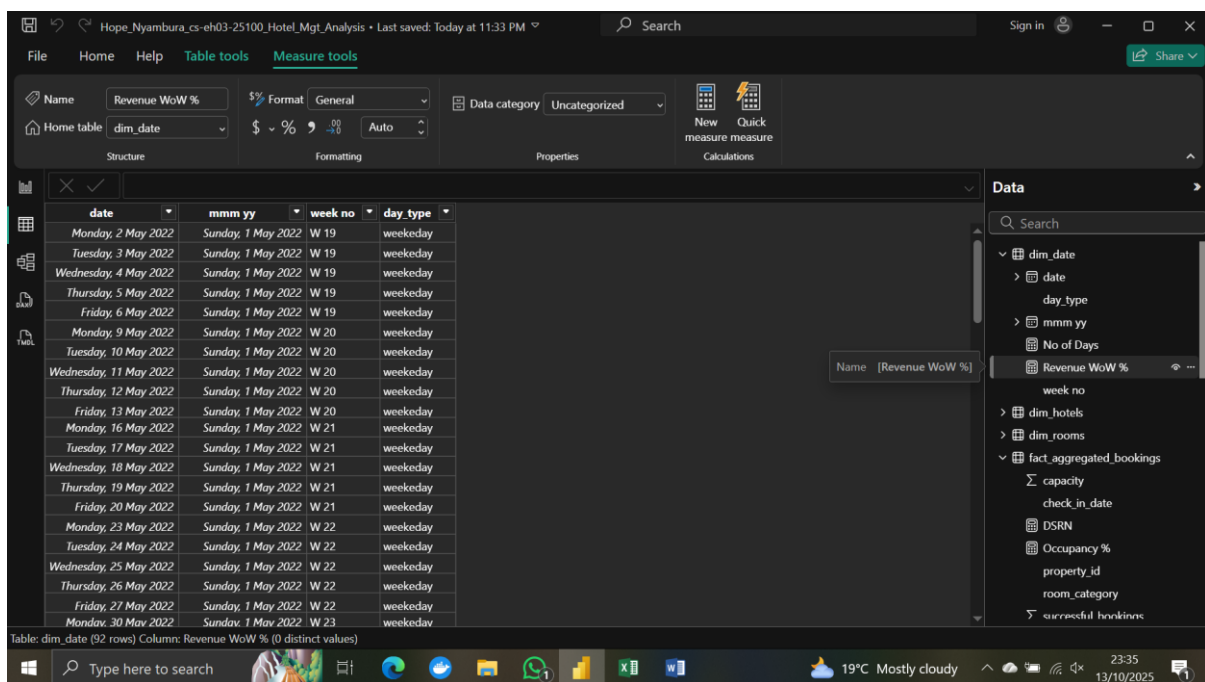
The second step was to load and prepare data for analysis. The dataset included several tables such as:

- dim_date (date and time details)
- dim_rooms (room types and categories)
- dim_hotels (hotel branch details)
- fact_aggregated_bookings
- fact_bookings (booking transactions and revenue data)

Using Power Query Editor, I performed the following transformations:

- Converted data types (dates, currency, and integers).
- Removed duplicate and null entries.
- Verified column consistency across all tables for correct relationships.

After cleaning, the data was loaded into the Power BI data model.

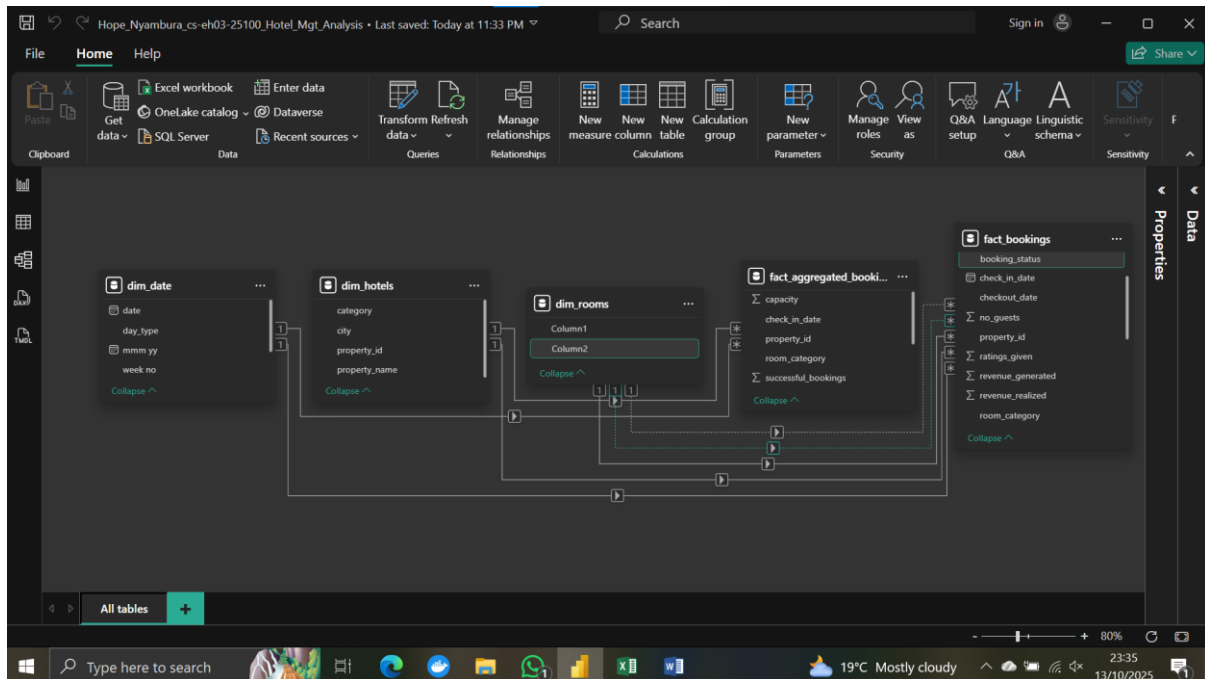


date	mmm yy	week no	day_type
Monday, 2 May 2022	Sunday, 1 May 2022	W 19	weekeday
Tuesday, 3 May 2022	Sunday, 1 May 2022	W 19	weekeday
Wednesday, 4 May 2022	Sunday, 1 May 2022	W 19	weekeday
Thursday, 5 May 2022	Sunday, 1 May 2022	W 19	weekeday
Friday, 6 May 2022	Sunday, 1 May 2022	W 19	weekeday
Monday, 9 May 2022	Sunday, 1 May 2022	W 20	weekeday
Tuesday, 10 May 2022	Sunday, 1 May 2022	W 20	weekeday
Wednesday, 11 May 2022	Sunday, 1 May 2022	W 20	weekeday
Thursday, 12 May 2022	Sunday, 1 May 2022	W 20	weekeday
Friday, 13 May 2022	Sunday, 1 May 2022	W 20	weekeday
Monday, 16 May 2022	Sunday, 1 May 2022	W 21	weekeday
Tuesday, 17 May 2022	Sunday, 1 May 2022	W 21	weekeday
Wednesday, 18 May 2022	Sunday, 1 May 2022	W 21	weekeday
Thursday, 19 May 2022	Sunday, 1 May 2022	W 21	weekeday
Friday, 20 May 2022	Sunday, 1 May 2022	W 21	weekeday
Monday, 23 May 2022	Sunday, 1 May 2022	W 22	weekeday
Tuesday, 24 May 2022	Sunday, 1 May 2022	W 22	weekeday
Wednesday, 25 May 2022	Sunday, 1 May 2022	W 22	weekeday
Thursday, 26 May 2022	Sunday, 1 May 2022	W 22	weekeday
Friday, 27 May 2022	Sunday, 1 May 2022	W 22	weekeday
Monday, 30 May 2022	Sunday, 1 May 2022	W 23	weekeday

Step 3: Build the Data Model (Star Schema)

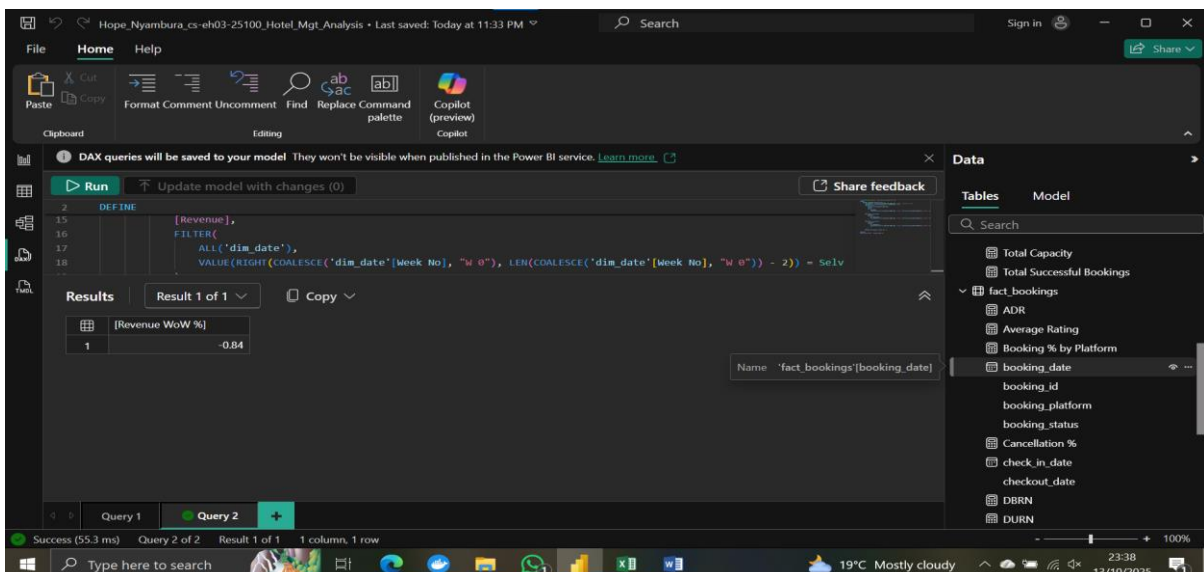
A Star Schema was created to structure the relationships between tables for efficient analysis.

- The fact_bookings table was placed at the center as the fact table containing revenue, booking status, and dates.
- The surrounding dimension tables (dim_rooms, dim_hotels and dim_date) were linked through their respective keys.



Step 4: Create DAX Calculations

To enhance the analytical capability of the dashboard, I created several DAX measures. These measures calculated performance metrics that were not directly available in the dataset.



DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more.](#)

Run Update model with changes (0)

```

2  DEFINE
15  [Revenue],
16  FILTER(
17  ALL('dim_date'),
18  VALUE(RIGHT(COALESCE('dim_date'[Week No], "W 0"), LEN(COALESCE('dim_date'[Week No], "W 0")) - 2)) = Solv

```

Results Result 1 of 1 Copy

	[Revenue WoW %]
1	-0.84

Name 'fact_bookings'[booking_date]

Tables Model

Search

- Total Capacity
- Total Successful Bookings
- fact_bookings
 - ADR
 - Average Rating
 - Booking % by Platform
 - booking_date
 - booking_id
 - booking_platform
 - booking_status
 - Cancellation %
 - check_in_date
 - checkout_date
 - DBRN
 - DURN

Success (55.3 ms) Query 2 of 2 Result 1 of 1 1 column, 1 row

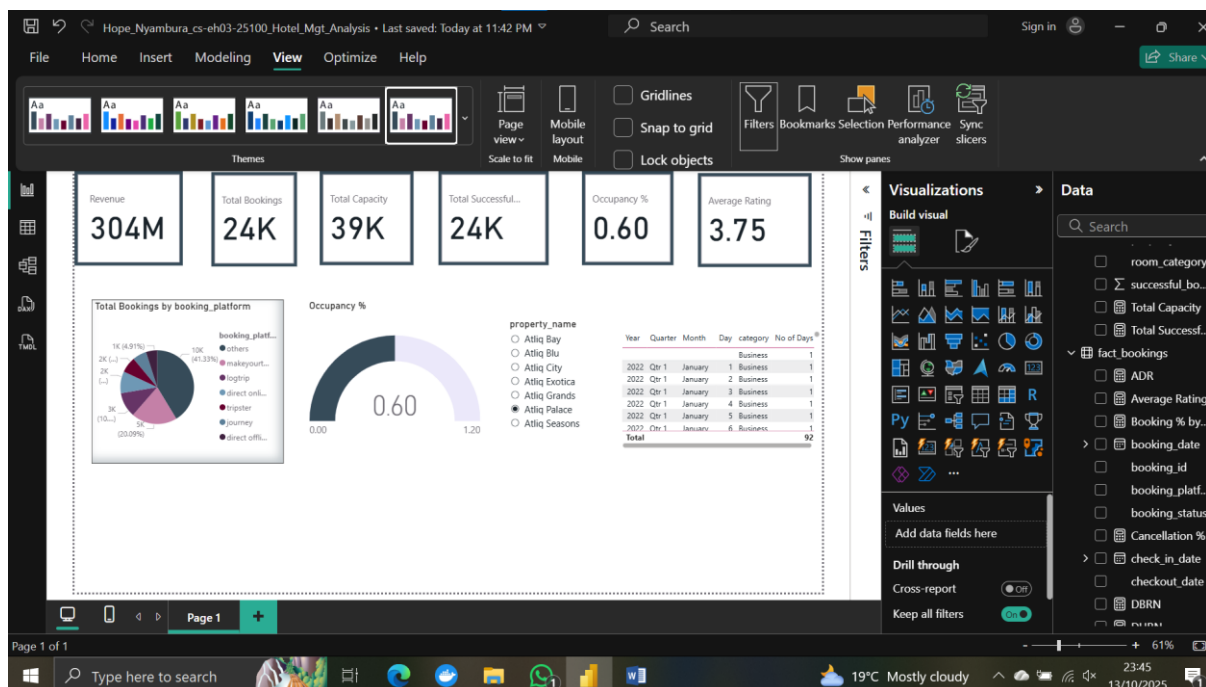
Step 5: Build Interactive Dashboards

After preparing the model and DAX measures, I created an interactive Power BI dashboard with multiple visuals to display hotel performance metrics.

The dashboard included:

- **KPI Cards:** showing Total Revenue, Occupancy Rate, and Cancellation Rate.
- **Donut Chart:** showing booking status distribution (Confirmed, Cancelled, Pending).
- **Table Visual:** listing top customers by revenue.
- **Slicers:** allowing filters by Hotel, Year, and Room Type.

This dashboard provided a clear, easy-to-navigate interface for management to make data-driven decisions.



Step 6: Publish and Share the Dashboard

Once the dashboard design was complete, I published it to the Power BI Service (Cloud) using the built-in *Publish* feature.

Steps followed:

1. Clicked File → Publish → To Power BI Service.
2. Chose My Workspace as the destination.
3. Verified the report online and used Publish to Web to create a public link.
4. Embedded the dashboard link in my portfolio website under *Business Intelligence Projects*.

This step demonstrated the complete lifecycle of a BI project from raw data to a live interactive report accessible via the web.

4. Conclusion

This assignment provided valuable hands-on experience in Business Intelligence using Power BI. I learned how to transform raw hotel data into a structured analytical model, create meaningful DAX measures, and visualize insights through an interactive dashboard.

The project strengthened my understanding of how BI tools help businesses monitor performance, detect trends, and improve decision-making. Publishing the dashboard online demonstrated my ability to deliver professional analytics solutions end-to-end, a skill essential for data analyst roles in real-world organizations.

5. Link to Published Power BI Dashboard

[\[https://drive.google.com/file/d/1tx9d-DY7magXOqoXWlnEvhQfyCZo29jJ/view?usp=sharing\]](https://drive.google.com/file/d/1tx9d-DY7magXOqoXWlnEvhQfyCZo29jJ/view?usp=sharing)